

K. Al Quds

time limit per test: 5 seconds
memory limit per test: 1024 megabytes

Al-Quds, known in English as Jerusalem, is one of the most ancient and significant cities in the world, with a history that spans over 5,000 years. Situated at the heart of the Middle East, it holds profound religious, cultural, and political significance for millions of people around the globe. For Jews, Al-Quds is home to the Western Wall, a remnant of the Second Temple and the holiest site in Judaism. For Christians, it is the city where Jesus Christ was crucified and resurrected, with the Church of the Holy Sepulchre standing as a monumental testament to these pivotal events in the Christian faith. For Muslims, Al-Quds houses the Al-Aqsa Mosque and the Dome of the Rock, which are considered the third holiest sites in Islam after Mecca and Medina.

Gohary, accompanied by his loyal woodpecker friend, is enjoying a peaceful stroll amidst the branches of a magnificent tree of n nodes each node has a worm in it. Suddenly, worms launch an attack, swarming the tree! Gohary's heart sinks as he realizes his woodpecker's beloved baby is perched on a distant node, far from the current location.

Driven by a sense of urgency and unwavering love for his feathered companion, Gohary vows to assist the woodpecker in reaching its offspring unharmed. However, the woodpecker has a limitation: it can only fend off worms with size at most k . To ensure a safe passage for the woodpecker, Gohary must strategically eliminate remaining worms along the path leading to the baby woodpecker's location.

Given q queries, you have to determine the number of worms Gohary needs to eliminate to clear a path for the woodpecker, enabling it to reach its baby as swiftly as possible.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains two integers n and q ($1 \leq n, q \leq 3 \cdot 10^5$) — nodes in the tree and the number of queries you have to answer.

The second line of each test case contains n integers. The i -th integer denotes the size of the worm of the i -th node.

It is guaranteed that the maximum size of each worm does not exceed n .

Each line of the next $n - 1$ lines contains two integers u and v ($1 \leq u, v \leq n$), indicating that there is an edge between node u and v in the tree. It is guaranteed that the given graph is a tree.

The next q lines contain three integers $u \ v \ k$ ($1 \leq u, v, k \leq n$) — the position of the woodpecker, the position of the baby woodpecker, and the size of the maximum worms the woodpecker can eat respectively.

It is guaranteed that the sum of values of n and q over all test cases does not exceed $3 \cdot 10^5$.

Output

For each query, print the number of worms Gohary needs to eliminate to clear a path for the woodpecker, enabling it to reach its baby as swiftly as possible.

Example

input	Copy
1 15 5 5 8 1 9 15 10 13 12 14 15 3 2 4 1 7 1 2 2 4 4 5 2 6 2 9 9 10 6 7 6 8 1 3 3 11 3 13 11 12 13 14 13 15 5 10 1 1 14 3 10 15 10 7 1 7 5 15 1	
output	Copy
5 2 2 3 6	

ACPC Scientific Committee Archive

Private

Participant

About Group

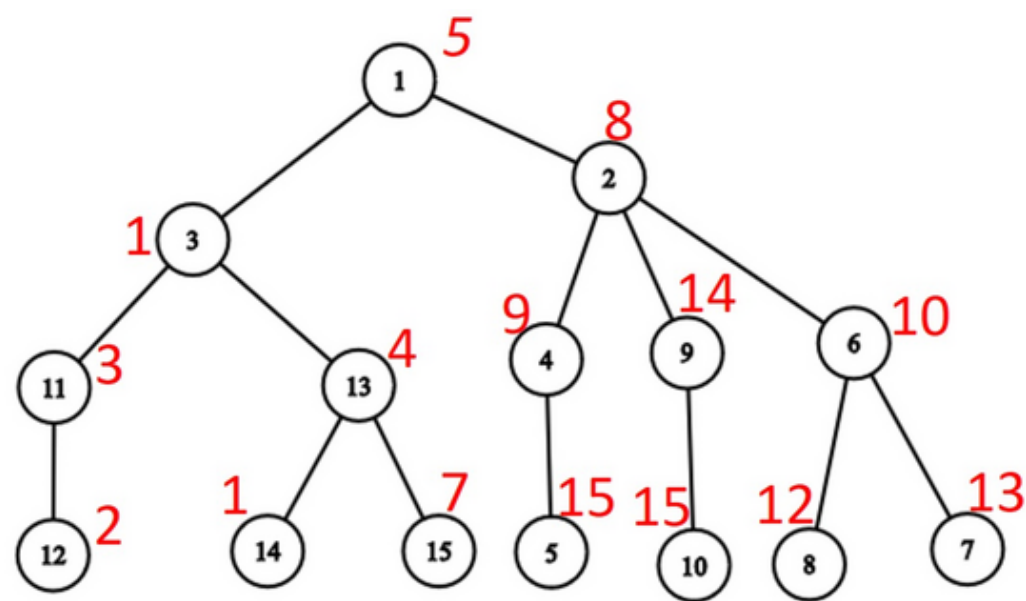
https://www.youtube.com/@MoamenAhmed98/play

Group website

Group Contests

- ECPC Qualifications 2025 Day 3 (Mirror)
- ECPC Qualifications 2025 Day 2 (Mirror)
- ECPC Qualifications 2025 Day 1 (Mirror)
- TCPC 2025
- ACPC Girls 2025
- ACPC Girls 2020
- HPV-CPC 2021
- Tunisian Collegiate Programming Contest (2017)
- Azhar Collegiate Programming Contest 2019
- The 2018 Menofia University Collegiate Programming Contest
- ICPC International Collegiate Programming Contest, Assiut Collegiate Programming Contest (2019)
- International Collegiate Programming Contest, GUC Collegiate Programming Contest (2019)
- International Collegiate Programming Contest, Aleppo University Collegiate Programming Contest(2018)
- The 2025 ICPC Egyptian Collegiate Programming Contest for Teens(Online Qualification)
- ECPC Qualifications 2021 - Day 4
- ACPC Kickoff 2025
- International Collegiate Programming Contest, Al-Baath, Tishreen, Aleppo and Damascus Collegiate Programming Contest (2021)
- The 2020 Egyptian Collegiate Programming Qualification Contest Day 2
- OCPC 2025 (Qualification)
- OCPC 2025
- ACPC 2024 (Mirror)
- LCPC 2024 (Mirror)
- JCPC 2024 (Mirror)
- EthCPC (Mirror)
- PCPC 2024
- LyCPC 2024
- AOCPC 2024
- AOCPC 2023
- QCPC 2024 (Mirror)

Note
The sample test case



- TCPC 2024 (Mirror)
- AICPC 2023
- SCPC 2024 (Mirror)
- SCPC 2024 For Teens (Mirror)
- ECPC 2024 (Mirror)
- ECPC Qualifications 2024 Day 6 (Mirror)
- ECPC Qualifications 2024 Day 5 (Mirror)
- ECPC Qualifications 2024 Day 4 (Mirror)
- ECPC Qualifications 2024 Day 3 (Mirror)
- ECPC Qualifications 2024 Day 2 (Mirror)
- ECPC Qualifications 2024 Day 1 (Mirror)
- International Collegiate Programming Contest, Damascus University Collegiate Programming Contest(2018)
- International Collegiate Programming Contest, Al-Baath, Tishreen, Aleppo and Damascus Collegiate Programming Contest (2021)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 2)
- ACPG Girls 2024
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 6)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 1)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 3)
- ACPG KickOff 2024
- SCPC 2023
- SCPC 2022
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 9)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 8)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 7)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 5)
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications Day 4)
- ACPG KickOff 2022
- ACPG KickOff 2023
- SCPC 2021
- SCPC 2020
- SCPC 2017
- SCPC 2019
- ECPC Teens 2024 (online qualification)
- Kickoff 2021
- ECPC 2021
- ACPG 2023
- Sa3edyCPC 2019
- The 2023 ICPC Egyptian Collegiate Programming Contest (Qualifications - Day 10)
- JCPC 2023